

1 Normalized Background Equilibrium State

$$\begin{aligned}
\vec{V}_0(x) &= V_{0z}(x)\hat{z} \\
Q_0^a &= \delta^{a1} \\
\partial_x \left(\int V_{0z}(x) dx \right) &= V_{0z}(x) \\
\vec{A}_0^1(x) &= A_{0z}^1(x)\hat{z} \\
\phi_0^1(x) &= - \int V_{0z}(x) \left(\int V_{0z}(x) dx \right) dx \\
E_{0x}^1 &= -\partial_x \phi_0^1 = V_{0z}(x) \int V_{0z}(x) dx \\
-\partial_x^2 A_{0z}^1(x) &= V_{0z}(x) \\
A_{0z}^1(x) &= \int \int V_{0z}(x) dx^2
\end{aligned}$$

Normalized Linearized Equations

Applying the re-normalized Yang-Mills magnetic field and Wong equations, we substitute $g \rightarrow \alpha_{YM}$, normalize the background color charge magnitude $Q \rightarrow 1$ (as $\hat{e}_Q$ is a unit vector), and absorb the speed of light $c \rightarrow 1$ with the perturbed scalar potential denoted as φ .

Perturbed Magnetic Field Components (b_i^a)

Evaluating the 9 components of the linearized magnetic field under the $a_x^a = 0$ gauge:

$$b_x^1 = -ik_z a_y^1 \tag{1}$$

$$b_x^2 = -ik_z a_y^2 - 2\alpha_{YM} A_{0z}^1 a_y^3 \tag{2}$$

$$b_x^3 = -ik_z a_y^3 + 2\alpha_{YM} A_{0z}^1 a_y^2 \tag{3}$$

$$b_y^1 = -\partial_x a_z^1 \tag{4}$$

$$b_y^2 = -\partial_x a_z^2 \tag{5}$$

$$b_y^3 = -\partial_x a_z^3 \tag{6}$$

$$b_z^1 = \partial_x a_y^1 \tag{7}$$

$$b_z^2 = \partial_x a_y^2 \tag{8}$$

$$b_z^3 = \partial_x a_y^3 \tag{9}$$

Perturbed Color Charge Components (q^a)

Utilizing the effective background coupling $\Phi_0^1 = V_{0z}A_{0z}^1$ and Doppler-shifted frequency $\Omega = \omega - k_z V_{0z}$, the 3 components of the color charge fluctuation are:

$$q^1 = 0 \quad (10)$$

$$q^2 = \frac{i\alpha_{YM}}{\Omega \left(1 - \frac{\alpha_{YM}^2 (\Phi_0^1)^2}{\Omega^2}\right)} \left((V_{0z}a_z^3 - \varphi^3) + \frac{i\alpha_{YM}\Phi_0^1}{\Omega} (V_{0z}a_z^2 - \varphi^2) \right) \quad (11)$$

$$q^3 = \frac{-i\alpha_{YM}}{\Omega \left(1 - \frac{\alpha_{YM}^2 (\Phi_0^1)^2}{\Omega^2}\right)} \left((V_{0z}a_z^2 - \varphi^2) - \frac{i\alpha_{YM}\Phi_0^1}{\Omega} (V_{0z}a_z^3 - \varphi^3) \right) \quad (12)$$

Normalized Linearized Ampere's Law

Applying the re-normalized parameters α_{YM} for the strong coupling constant, setting the speed of light $c \rightarrow 1$, normalizing the primary background charge magnitude $Q \rightarrow 1$, and operating in naturalized units where $4\pi n_0 \rightarrow 1$, we obtain the final linearized equations.

Primary Color Index ($a = 1$)

For the primary Abelian-like state, the inverted velocity fluctuations are fully decoupled from the background vector potentials:

$$v_x = -ik_z b_y^1 + i\omega e_x^1 \quad (13)$$

$$v_y = ik_z b_x^1 - \partial_x b_z^1 + i\omega e_y^1 \quad (14)$$

$$v_z = \partial_x b_y^1 + i\omega e_z^1 \quad (15)$$

Off-Diagonal Color Indices ($a = 2, 3$)

Expanding the generalized Ampere operator for the off-diagonal states, evaluating the structure constants $\epsilon^{231} = 1$ and $\epsilon^{321} = -1$, and decomposing into Cartesian components yields the fully coupled transverse and longitudinal constraints:

For the second color state ($a = 2$):

$$-ik_z b_y^2 + i\omega e_x^2 - \alpha_{YM} (-a_z^3 B_{0y}^1 - A_{0z}^1 b_y^3 + \phi_0^1 e_x^3 + \varphi^3 E_{0x}^1) = 0 \quad (16)$$

$$ik_z b_x^2 - \partial_x b_z^2 + i\omega e_y^2 - \alpha_{YM} (A_{0z}^1 b_x^3 + \phi_0^1 e_y^3) = 0 \quad (17)$$

$$\partial_x b_y^2 + i\omega e_z^2 - \alpha_{YM} \phi_0^1 e_z^3 = V_{0z} q^2 \quad (18)$$

For the third color state ($a = 3$):

$$-ik_z b_y^3 + i\omega e_x^3 + \alpha_{YM} (-a_z^2 B_{0y}^1 - A_{0z}^1 b_y^2 + \phi_0^1 e_x^2 + \varphi^2 E_{0x}^1) = 0 \quad (19)$$

$$ik_z b_x^3 - \partial_x b_z^3 + i\omega e_y^3 + \alpha_{YM} (A_{0z}^1 b_x^2 + \phi_0^1 e_y^2) = 0 \quad (20)$$

$$\partial_x b_y^3 + i\omega e_z^3 + \alpha_{YM} \phi_0^1 e_z^2 = V_{0z} q^3 \quad (21)$$

2 Final equations

$$(22) \quad [-\partial_x^2 - \omega^2 + k_z^2 \mp \alpha_{YM} k_z A_{0z}^1 - 2\alpha_{YM}^2 (A_{0z}^1)^2 + \alpha_{YM}^2 (\phi_0^1)^2] a_y^\pm = 0$$

$$(-\partial_x^2 - \omega^2 + k_z^2 + \alpha_{YM}^2 (\phi_0^1)^2 - \alpha_{YM}^2 (A_{0z}^1)^2) a_z^+ = \frac{\alpha_{YM}^2 V_{0z}}{\Omega + \alpha_{YM} \Phi_0^1} \left(V_{0z} - \frac{k_z + \alpha_{YM} A_{0z}^1}{\omega + \alpha_{YM} \phi_0^1} \right) a_z^+ \quad (23)$$

3 Step potential

$$\left[-\partial_x^2 - \omega^2 + k_z^2 \mp \alpha_{YM} k_z \left(\frac{v}{2} x^2 \right) - \alpha_{YM}^2 \frac{v^2}{2} x^4 + \alpha_{YM}^2 \frac{v^4}{4} x^4 \right] a_y^\pm = 0 \quad (24)$$

$$\left(-\partial_x^2 - \omega^2 + \alpha_{YM}^2 \frac{v^4}{4} x^4 \right) a_z^+ + \left(k_z - \alpha_{YM} \frac{v}{2} x^2 \right) \left(\omega + \alpha_{YM} \frac{v^2}{2} x^2 \right) \varphi^+ = \frac{\alpha_{YM}^2 V_{0z}}{\Omega + \alpha_{YM} \frac{v}{2} x^2 (V_{0z} - v)} (V_{0z} a_z^+ - \varphi^+) \quad (25)$$

Substitution and Simplification

We begin with the normalized constraint equation for the complex longitudinal fields, assuming the constant of integration vanishes for the physical perturbations. Applying the step potential substitutions $A_{0z}^1(x) = \frac{v}{2} x^2$ and $\phi_0^1(x) = \frac{v^2}{2} x^2$:

$$\left(k_z + \alpha_{YM} \frac{v}{2} x^2 \right) a_z^+ - \left(\omega + \alpha_{YM} \frac{v^2}{2} x^2 \right) \varphi^+ = 0 \quad (26)$$

Rearranging this to solve for φ^+ , we obtain:

$$\varphi^+ = \frac{k_z + \alpha_{YM} \frac{v}{2} x^2}{\omega + \alpha_{YM} \frac{v^2}{2} x^2} a_z^+ \quad (27)$$

Next, we substitute Equation (27) into the fully normalized z -component wave equation, explicitly utilizing the effective background coupling $\Phi_0^1(x) = \frac{v}{2}x^2(V_{0z} - v)$:

$$\begin{aligned} \left(-\partial_x^2 - \omega^2 + \alpha_{YM}^2 \frac{v^4}{4}x^4\right) a_z^+ + \left(k_z - \alpha_{YM} \frac{v}{2}x^2\right) \left(\omega + \alpha_{YM} \frac{v^2}{2}x^2\right) \varphi^+ \\ = \frac{\alpha_{YM}^2 V_{0z}}{\Omega + \alpha_{YM} \frac{v}{2}x^2(V_{0z} - v)} (V_{0z} a_z^+ - \varphi^+) \end{aligned}$$

Focusing on the left-hand side, the substitution yields a convenient cancellation of the $\left(\omega + \alpha_{YM} \frac{v^2}{2}x^2\right)$ terms:

$$\begin{aligned} \text{LHS} &= \left(-\partial_x^2 - \omega^2 + \alpha_{YM}^2 \frac{v^4}{4}x^4\right) a_z^+ + \left(k_z - \alpha_{YM} \frac{v}{2}x^2\right) \left(\omega + \alpha_{YM} \frac{v^2}{2}x^2\right) \left[\frac{k_z + \alpha_{YM} \frac{v}{2}x^2}{\omega + \alpha_{YM} \frac{v^2}{2}x^2} a_z^+\right] \\ &= \left(-\partial_x^2 - \omega^2 + \alpha_{YM}^2 \frac{v^4}{4}x^4\right) a_z^+ + \left(k_z - \alpha_{YM} \frac{v}{2}x^2\right) \left(k_z + \alpha_{YM} \frac{v}{2}x^2\right) a_z^+ \\ &= \left(-\partial_x^2 - \omega^2 + k_z^2 + \alpha_{YM}^2 \frac{v^4}{4}x^4 - \alpha_{YM}^2 \frac{v^2}{4}x^4\right) a_z^+ \end{aligned}$$

Applying the same substitution to the right-hand side and factoring out a_z^+ gives:

$$\text{RHS} = \frac{\alpha_{YM}^2 V_{0z}}{\Omega + \alpha_{YM} \frac{v}{2}x^2(V_{0z} - v)} \left(V_{0z} - \frac{k_z + \alpha_{YM} \frac{v}{2}x^2}{\omega + \alpha_{YM} \frac{v^2}{2}x^2}\right) a_z^+$$

Equating the simplified LHS and RHS, we arrive at the final decoupled equation for a_z^+ strictly in terms of the variable x :

$$\left(-\partial_x^2 - \omega^2 + k_z^2 + \frac{\alpha_{YM}^2 v^2}{4}x^4(v^2 - 1)\right) a_z^+ = \frac{\alpha_{YM}^2 V_{0z}}{\Omega + \alpha_{YM} \frac{v}{2}x^2(V_{0z} - v)} \left(V_{0z} - \frac{k_z + \alpha_{YM} \frac{v}{2}x^2}{\omega + \alpha_{YM} \frac{v^2}{2}x^2}\right) a_z^+ \quad (28)$$